Digital Literacy

Elliot Attipoe

Emerging Technologies

Emerging Technologies

- Artificial Intelligence
- 3D printing
- Augmented Reality/Virtual Reality
- Internet of Things (IoT)
- 5G/6G networks
- Drones Advancement
- Biometrics
- Blockchain

Artificial Intelligence

1. Artificial Intelligence is a method of making a computer, a computer-controlled robot, or a software think intelligently like the human mind.

2. Artificial Intelligence is the theory and development of computer systems able to perform tasks normally requiring human intelligence, such as visual perception, speech recognition, decision-making, and translation between languages.

Artificial Intelligence Applications in 2023

- E-Commerce
- Lifestyle
- Robotics
- Human Resource
- Social Media

AI Application in E-Commerce

- Personalized Shopping
- Virtual shopping assistants and chatbots
- Fraud Prevention

Applications of AI in Lifestyle

- Spam Filters
- Facial Recognition
- Recommendation System
- Virtual assistants
- Autonomous Vehicles

Virtual Assistants

- AI is to understand many natural language requests in other ways than rudimentary commands.
- Common examples of virtual assistants:
  - –Apple's Siri
  - –Amazon's Alexa
  - –OpenAI's ChatGPT

Autonomous Cars

- Autonomous Car (driverless **car**, self-driving **car**, robotic **car**) is a **vehicle** that is capable of sensing its environment and navigating without human input.)





Corporations working on autonomous vehicles



CBINSIGHTS

www.cbinsights.com

Robotaxis (Driverless Taxi)

Robotaxis: the future of transport

Robotaxis Roaming Around Yokohama Are Winning Over Unlikely Fans

Waymo's driverless robotaxis are now doing airport trips in Phoenix

San Francisco opposes robotaxi expansion due to erratic behavior of vehicles

Baidu starts offering nighttime driverless taxis in China

Rita Liao @ritacyliao / 11:16 PM UTC-8 • December 26, 2022

Waymo expands robotaxi service to LA

The company's operations are currently focused on San Francisco and Phoenix, Arizona

Applications of AI in Robotics

- Carrying goods in hospitals, factories, and warehouses
- Cleaning offices and large equipment
- Inventory management
- Assembly plants

Artificial Intelligence (Robots)

AI and robots could threaten your career within 5 years

Marguerite Ward | @forwardist | 11:42 AM ET Thu, 5 Oct 2017

Artificial intelligence may replace 40% of all jobs: Bridgewater founder Ray Dalio

Published September 22, 2017 | Business Leaders | FOXBusiness

Artificial intelligence will replace half of all jobs in the next decade, says widely followed technologist

Sophia Yan

Published 11:30 AM ET Thu, 27 April 2017 | Updated 8:49 PM ET Thu, 27 April 2017

CNBC

50% of low-skilled jobs will be replaced by AI and automation, report claims

A report from Ball State University found that machines will master half of low-skilled jobs currently performed by humans, and a quarter of those jobs are at risk of being sent offshore.

By Hope Reese | July 21, 2017, 8:47 AM PST

Artificial Intelligence (Robots)

30 companies already replacing humans with robots

Japanese insurance firm replaces 34 staff with AI

5 January 2017 | Asia

Share

Robots are replacing holiday workers at Amazon with the online retailer hiring 20,000 fewer staff to tackle this year's Christmas rush, says analyst

Capita to replace staff with robots to save money

Outsourcing giant to axe 2,000 jobs and use 'proprietary robotic solutions' after clients cut spending following Brexit vote

Application of AI in Human Resource

- Artificial Intelligence helps with blind hiring.

- AI systems can scan job candidates' profiles, and resumes to provide recruiters an understanding of the talent pool they must choose from.

This robot could be conducting your next job interview



This eerie device could be the future of job interviews. Image: FURHAT ROBOTICS

Tengai

Applications of AI in Social Media

- **Instagram**
  - AI considers your likes and the accounts you follow to determine what posts you are shown on your explore tab.

- **Facebook**
  - Artificial Intelligence is also used along with a tool called DeepText. With this tool, Facebook can understand conversations better. It can be used to translate posts from different languages automatically.

Applications of AI in Social Media

- **Twitter**
  - AI is used by Twitter for fraud detection, removing propaganda, and hateful content. Twitter also uses AI to recommend tweets that users might enjoy, based on what type of tweets they engage with.

- **TikTok**
  - Tiktok uses AI-based recommendation algorithm to drive user engagement. The app recommends videos based on a users interest.
  - TikTok's AI tracks each and every behavior as users watch the recommended videos. (emotions such as sadness, happiness, apprehension etc)

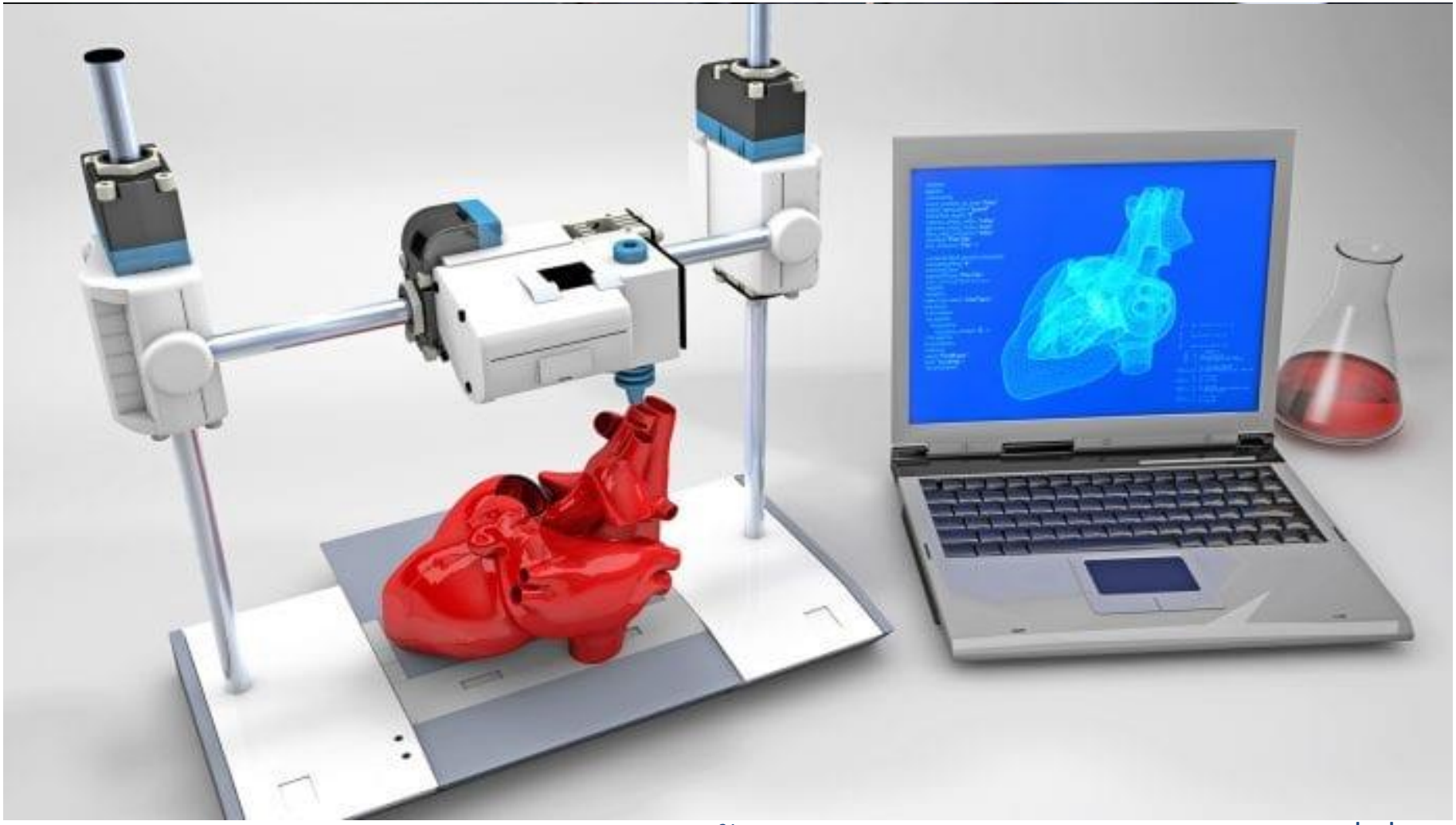
Fully Digital Bank

- A bank without branches.
- All banking services are online – gray area in Ghana.
- **Stats**
- A Bankrate survey shows that a third of Americans haven't visited their banks in the last six months.
- The British bankers' association says that nearly £1bn is transferred daily using the Internet.

3D printing

- 3D printing is also known as additive manufacturing.
- 3D printing is a process of making three dimensional solid objects from a digital file.
- Applications of 3D printers
  - Consumer products (eyewear, footwear, furniture)
  - Dental products
  - Prosthetics
  - Body parts like the ear and nose

3D Printing





chinese company 3D prints 10 recycled concrete houses in 24 hours

Emerging Technologies

- Artificial Intelligence
- 3D printing
- Augmented Reality/Virtual Reality
- Internet of Things (IoT)
- 5G/6G networks
- Drones Advancement
- Biometrics
- Blockchain

FIGURE 1 | **Employment trends for jobs in the United States at high risk of automation, 2007–2018**

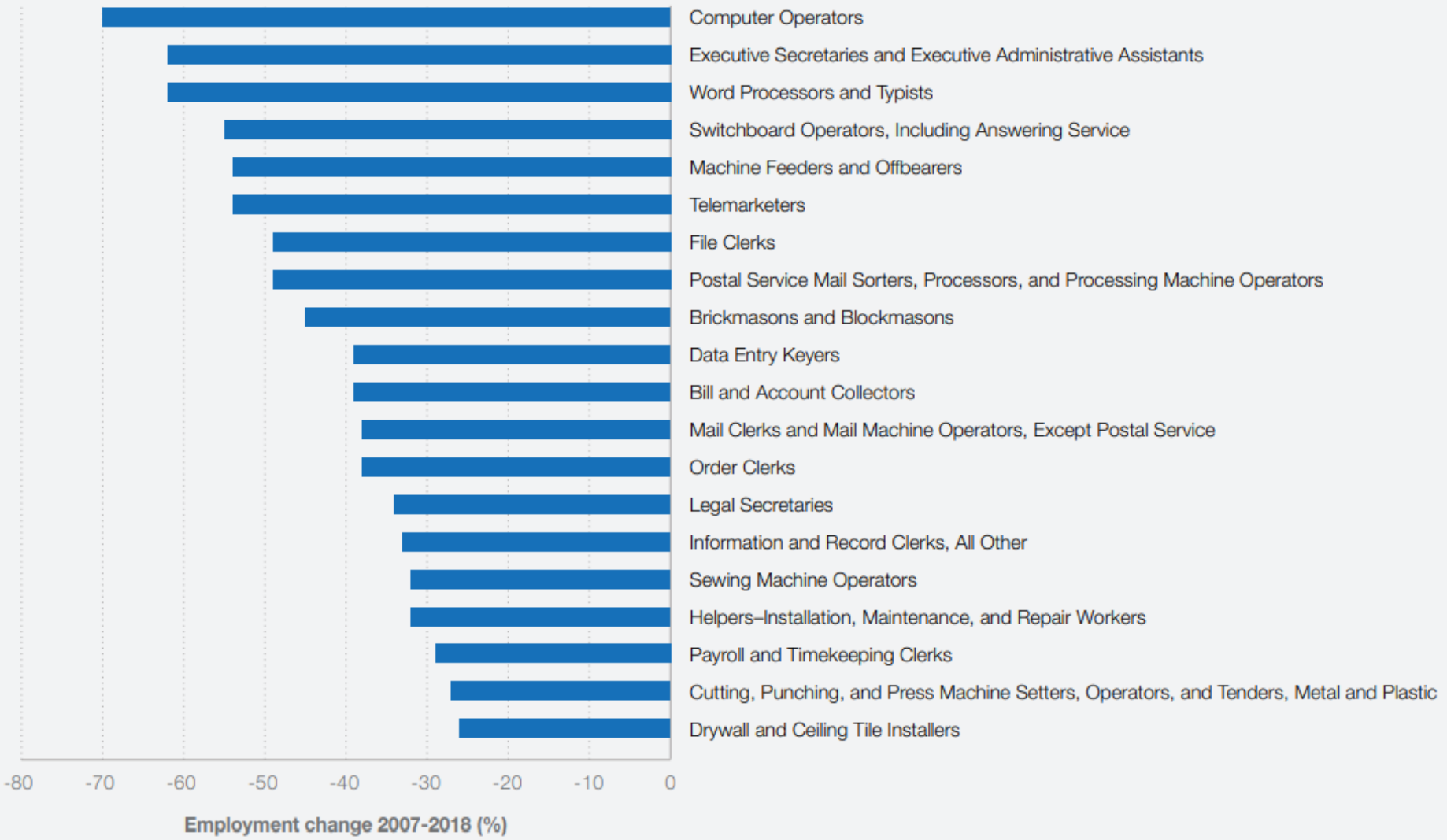


FIGURE 18

Technologies likely to be adopted by 2025 (by share of companies surveyed)

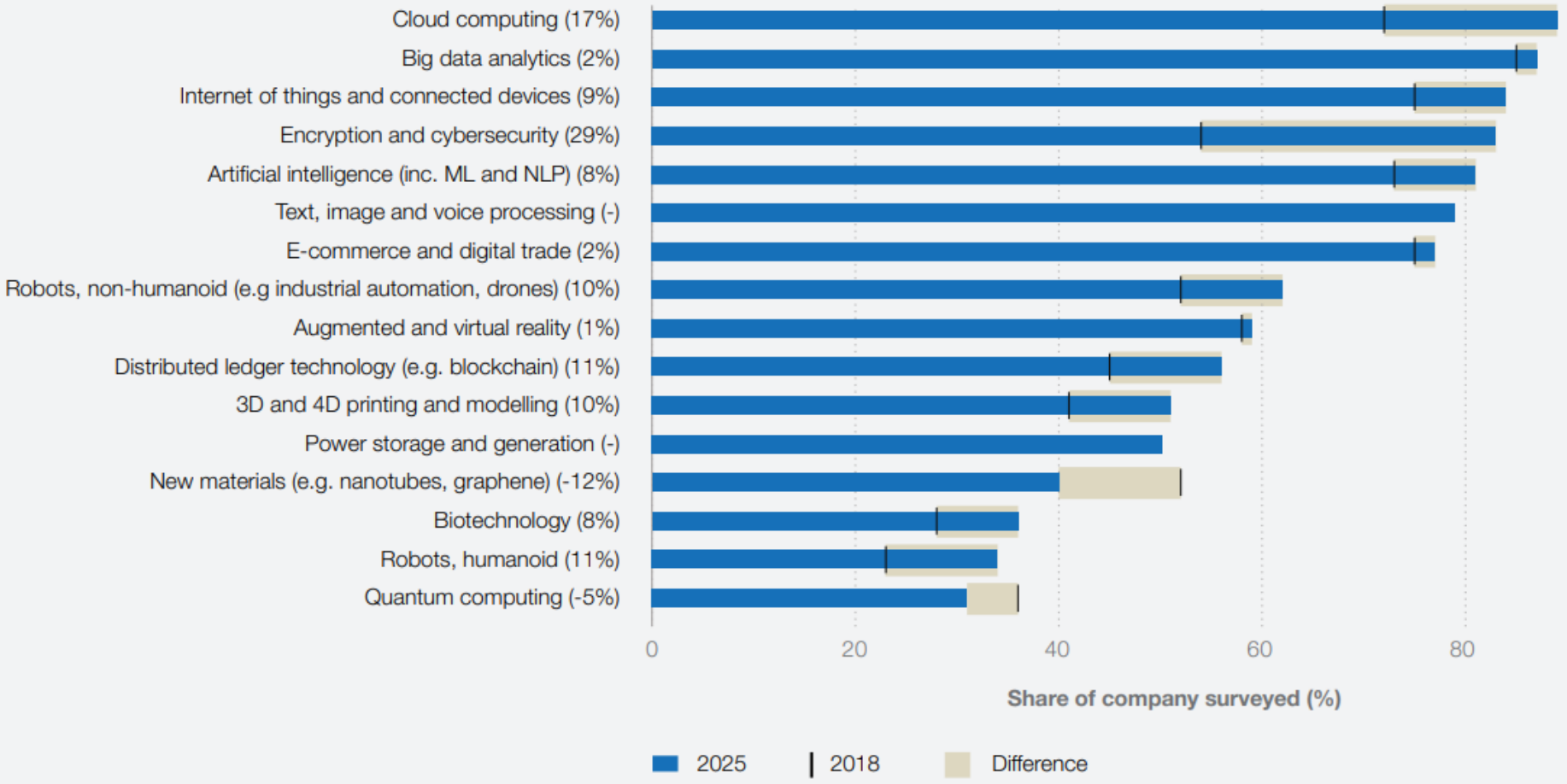


FIGURE 22

Top 20 job roles in increasing and decreasing demand across industries

↗ Increasing demand

1	Data Analysts and Scientists
2	AI and Machine Learning Specialists
3	Big Data Specialists
4	Digital Marketing and Strategy Specialists
5	Process Automation Specialists
6	Business Development Professionals
7	Digital Transformation Specialists
8	Information Security Analysts
9	Software and Applications Developers
10	Internet of Things Specialists
11	Project Managers
12	Business Services and Administration Managers
13	Database and Network Professionals
14	Robotics Engineers
15	Strategic Advisors
16	Management and Organization Analysts
17	FinTech Engineers
18	Mechanics and Machinery Repairers
19	Organizational Development Specialists
20	Risk Management Specialists

↘ Decreasing demand

1	Data Entry Clerks
2	Administrative and Executive Secretaries
3	Accounting, Bookkeeping and Payroll Clerks
4	Accountants and Auditors
5	Assembly and Factory Workers
6	Business Services and Administration Managers
7	Client Information and Customer Service Workers
8	General and Operations Managers
9	Mechanics and Machinery Repairers
10	Material-Recording and Stock-Keeping Clerks
11	Financial Analysts
12	Postal Service Clerks
13	Sales Rep., Wholesale and Manuf., Tech. and Sci.Products
14	Relationship Managers
15	Bank Tellers and Related Clerks
16	Door-To-Door Sales, News and Street Vendors
17	Electronics and Telecoms Installers and Repairers
18	Human Resources Specialists
19	Training and Development Specialists
20	Construction Laborers

Digital Literacy

Digital Literacy

- **Digital literacy** involves having a current knowledge and understanding of **computers, mobile devices**, the **Internet**, and **related technologies**.

What is a computer?

- An electronic device that processes data into information that is useful to people.

Core functions of a computer

i. It gathers data (or allows users to **input** data).
ii. It **processes** that data into information.
iii. It **outputs** data or information.
iv. It **stores** data and information.
v. It **communicates** with other devices (Additional functionality)

Types of Computers

1. Personal computers

- Computers designed for only one person to use at a time

2. Multi-user computers

- Computers designed to serve group of people

Personal Computers (PC)

- Desktop PC
- Notebook PC (Laptop)
- Tablet PC
- Smartphone

Smartphone







Smartphone can be defined as a portable computer device that combines mobile telephone and computing functions into one unit.

Smartphone - OS



Mobile Operating Systems (Alternatives to Android and iOS)

- SIRIN OS
- KaiOS
- Ubuntu Touch
- Tizen OS
- Harmony OS
- LineaseOS
- Paranoid Android

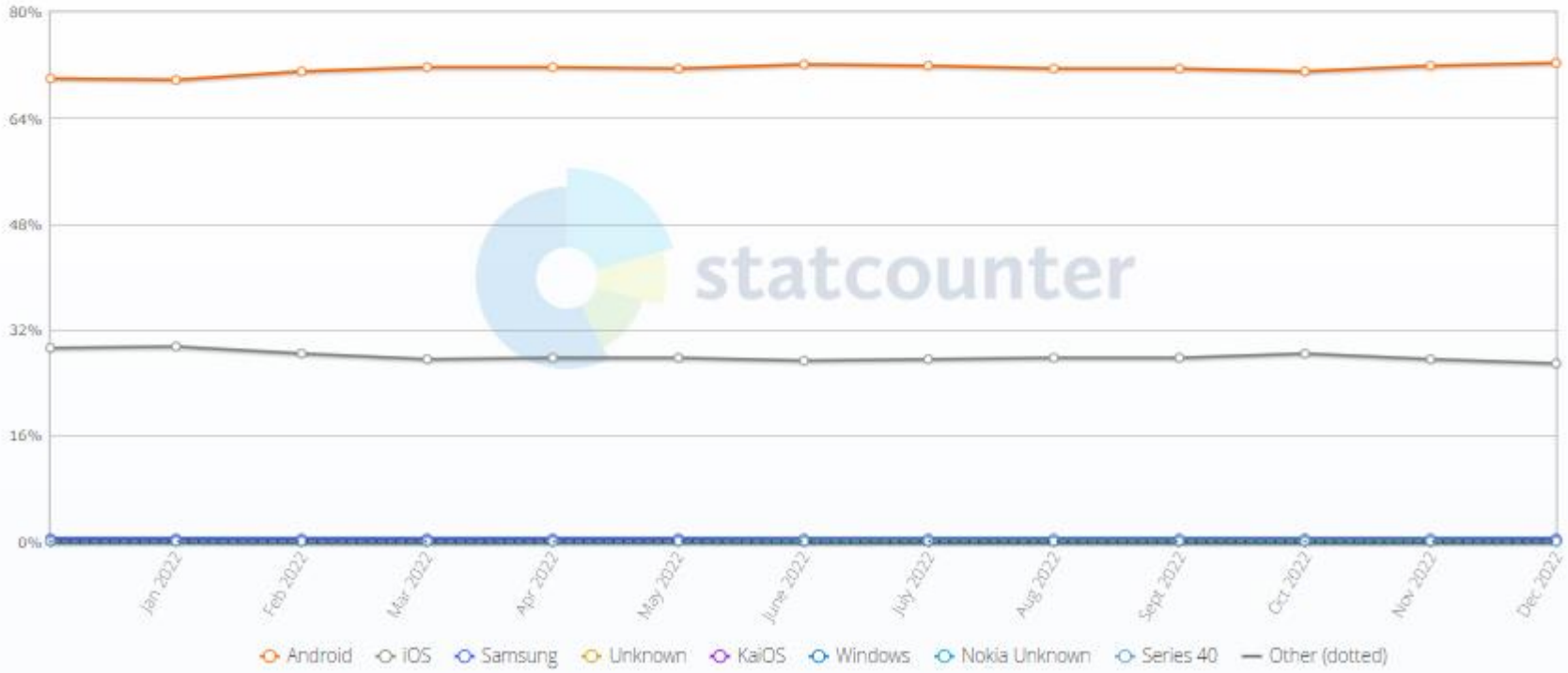
Android	iOS	Samsung	Unknown	KaiOS	Windows
72.37%	26.98%	0.36%	0.13%	0.11%	0.02%

Mobile Operating System Market Share Worldwide - December 2022

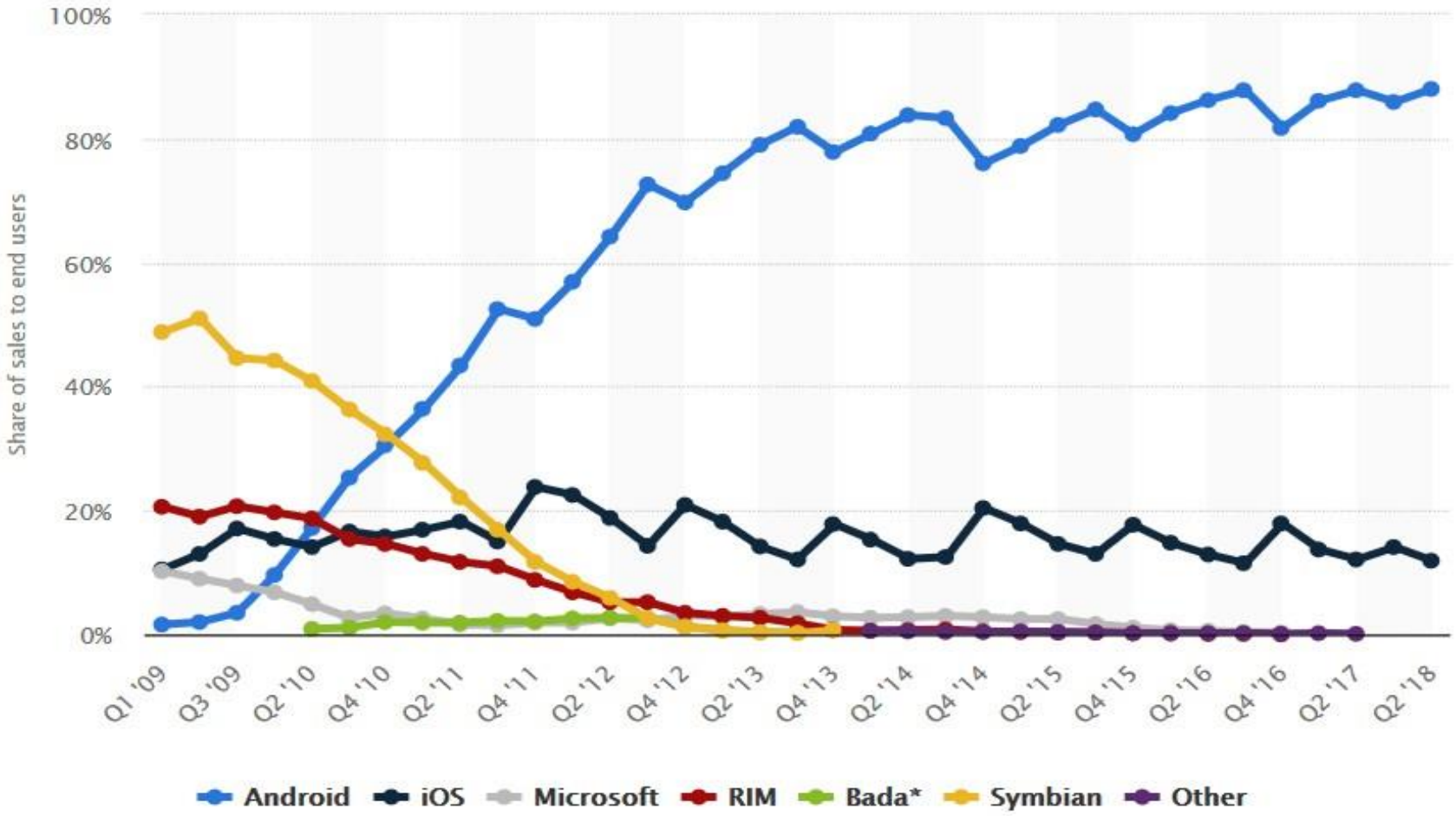
Mobile Operating System Market Share Worldwide

Dec 2021 - Dec 2022

Edit Chart Data



Mobile OS


100%
80%
60%
40%
20%
0%
Share of sales to end users
Q1 '09
Q3 '09
Q2 '10
Q4 '10
Q2 '11
Q4 '11
Q2 '12
Q4 '12
Q2 '13
Q4 '13
Q2 '14
Q4 '14
Q2 '15
Q4 '15
Q2 '16
Q4 '16
Q2 '17
Q2 '18
Android
iOS
Microsoft
RIM
Bada*
Symbian
Other

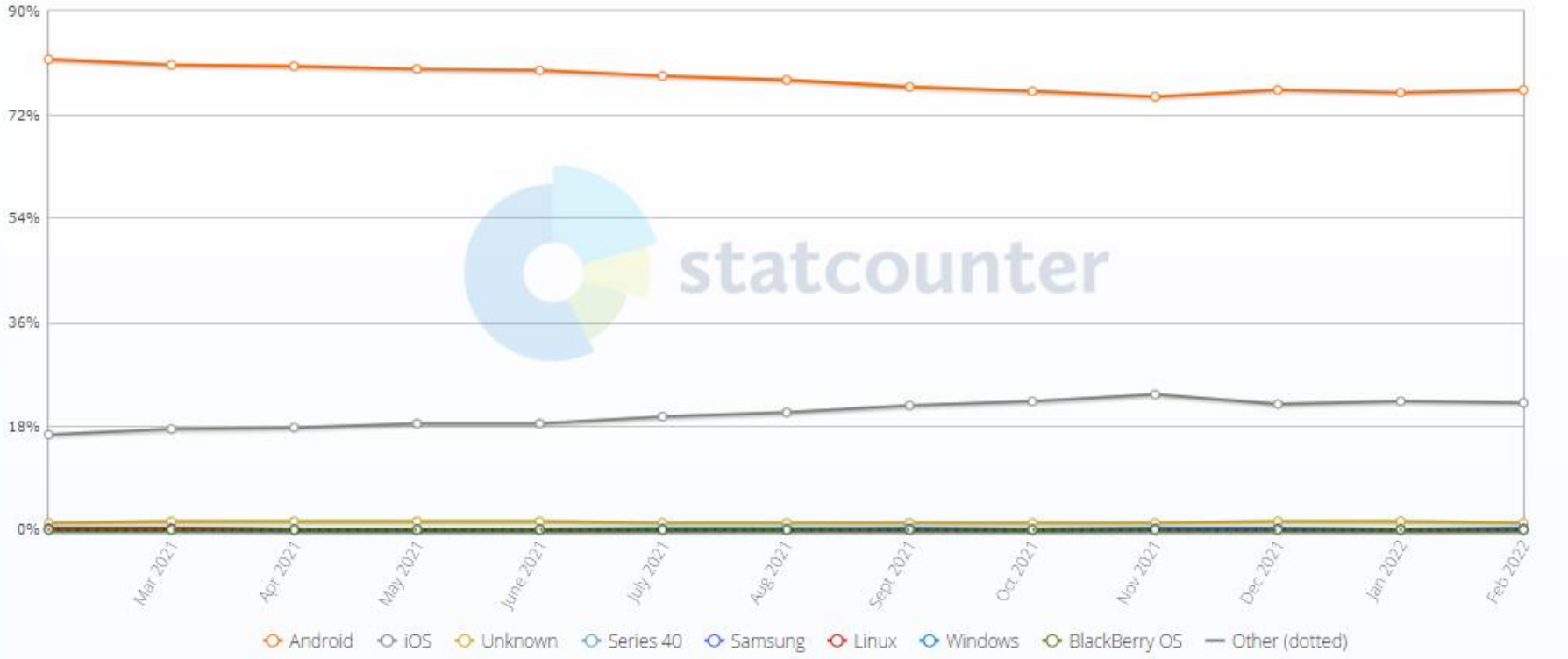
Android	iOS	Unknown	Samsung	Series 40	Windows
76.4%	22.07%	1.15%	0.12%	0.06%	0.05%

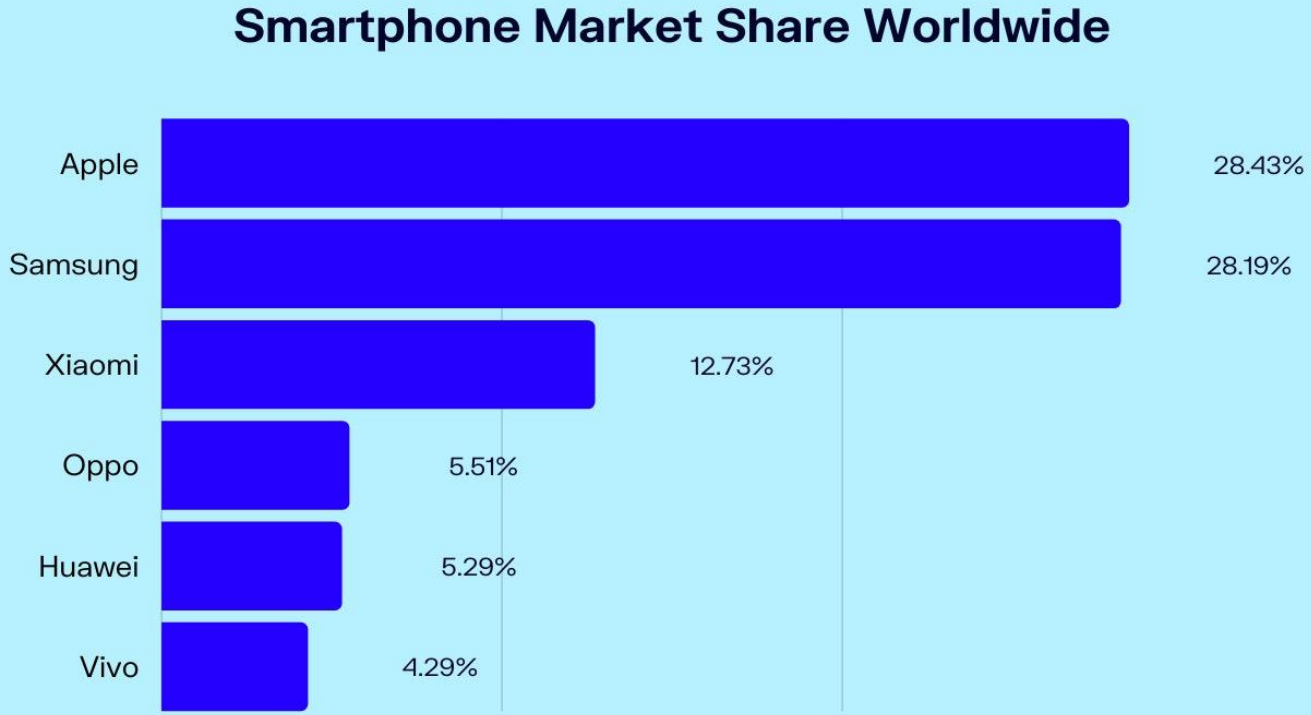
Mobile Operating System Market Share in Ghana - February 2022

Mobile Operating System Market Share Ghana

Feb 2021 - Feb 2022

Edit Chart Data



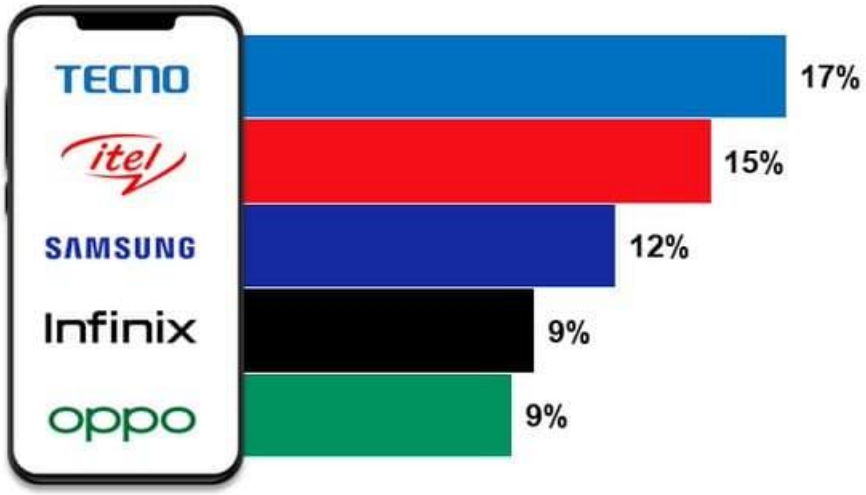


Source: GS Statcounter

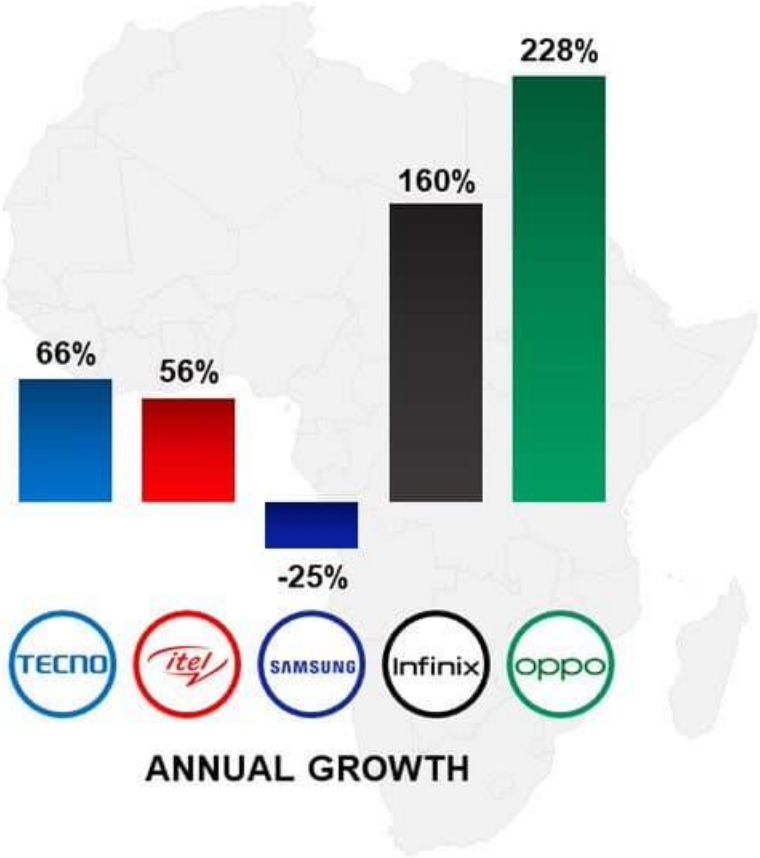
OBERLO

Counterpoint
Technology Market Research

Top smartphone vendors Q2 2021 | Africa



SALES SHARE



ANNUAL GROWTH

Source: Counterpoint Research Market Pulse service

Smartphone Apps



Laptop (Notebook)



Windows based laptop



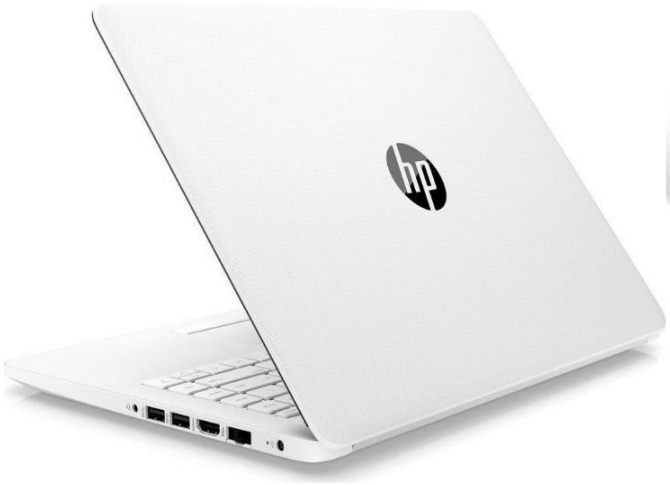
Apple's Macbook

Windows based PCs









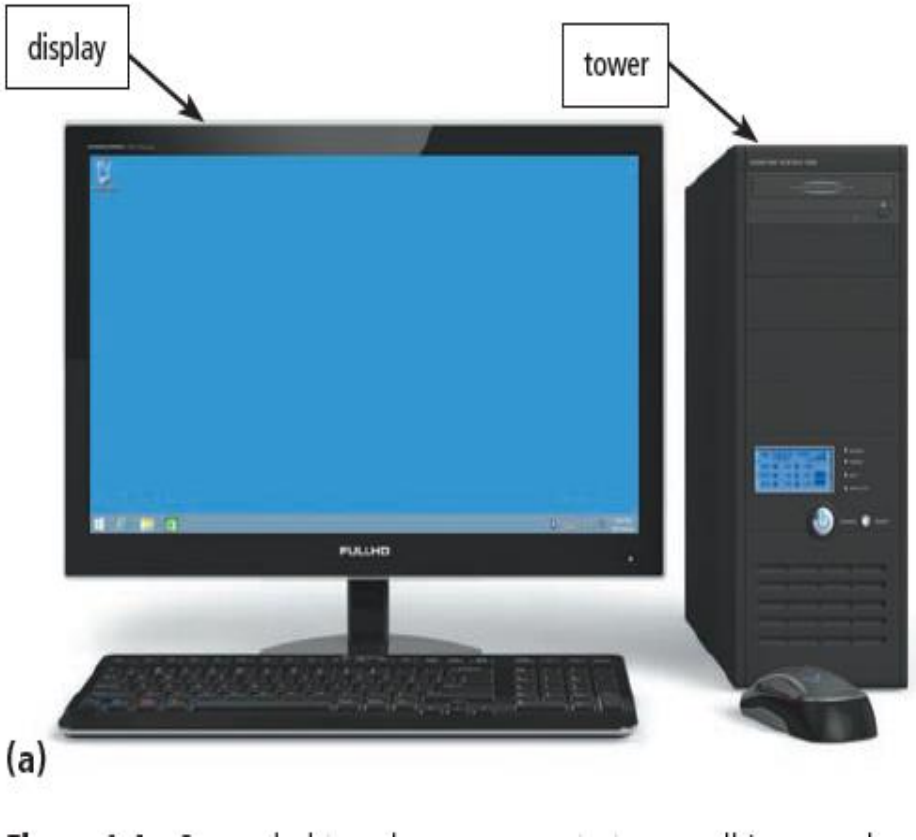


Apple – MacBook





Desktop PC





All-in-one Desktop PC



HP All in One 22 (1G9U0EA 22-...

GHS 6,980.00

Goodluck Africa



Lenovo T24t-20 23.8" 16:9 Multi-...

GHS 5,672.67

B&H Photo-Video-...

$459.99 + tax



Lenovo ThinkCentre Tiny-In-One 27"...

GHS 5,352.15

B&H Photo-Video-...

$434.00 + tax

Tablets


Library
Books (9)
Magazines (5)
Apps (30)
Google
MY LIBRARY
THE HUNGER GAMES
TRANSFORMERS
Video Hub
Game Hub
Music Hub
SAMSUNG

Personal computers (PC)

- Desktop PC
- Notebook PC (Laptop)
- Tablet PC
- Smartphone

Laptop Brands – which brand is the best?



Top 10 laptop brands

















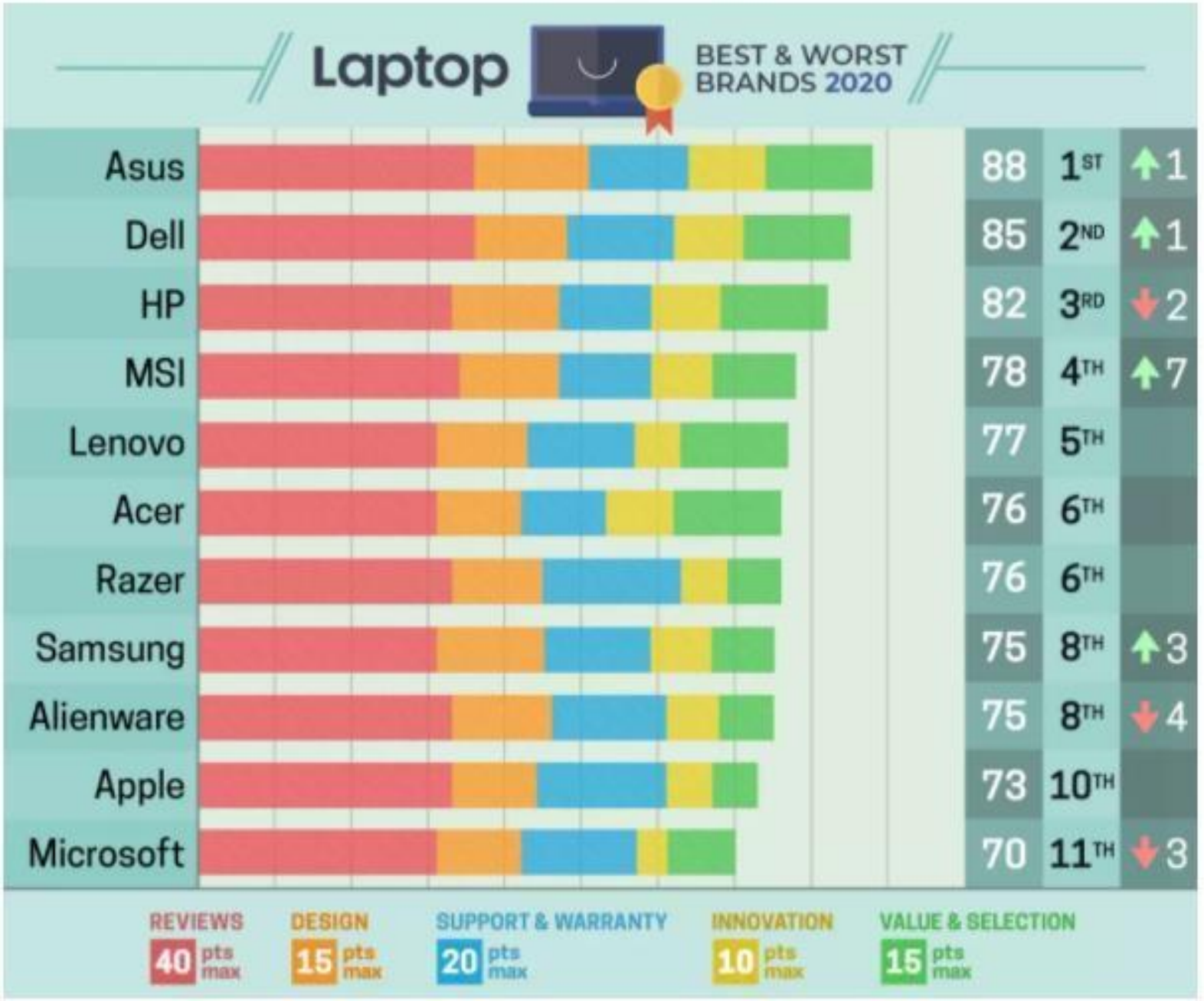




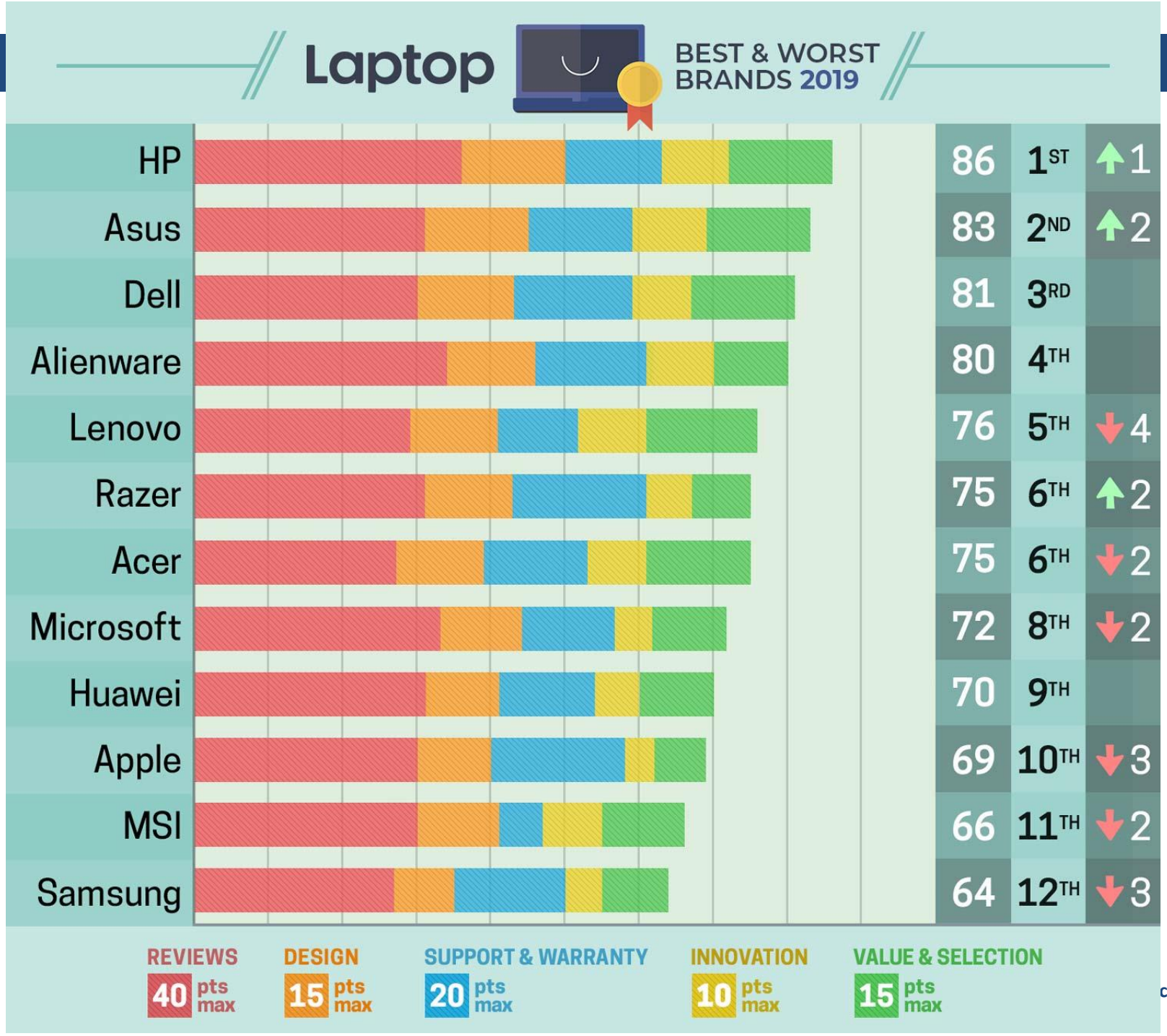




Source: https://mylaptopguide.com/best-laptop-brands/



(Image credit: Laptop Mag)


Laptop
BEST & WORST BRANDS 2019
HP 86 1ST ↑1
Asus 83 2ND ↑2
Dell 81 3RD
Alienware 80 4TH
Lenovo 76 5TH ↓4
Razer 75 6TH ↑2
Acer 75 6TH ↓2
Microsoft 72 8TH ↓2
Huawei 70 9TH
Apple 69 10TH ↓3
MSI 66 11TH ↓2
Samsung 64 12TH ↓3
REVIEWS 40 pts max
DESIGN 15 pts max
SUPPORT & WARRANTY 20 pts max
INNOVATION 10 pts max
VALUE & SELECTION 15 pts max

Factors to consider when buying a computer

1. Purpose or use
2. Brand
3. Price (budget)
4. Operating System type (Windows, Macintosh, Chromebook, Linux)
5. Specs (Memory, storage (SSD, HDD), Processor, Screen Size, weight, colour etc)
6. Battery Life
7. After sale services (warranty etc)

Factors to consider when buying a smartphone

- **Battery** – measured in mAh (Milliamp Hours).
- **Memory** – one of the factors that determine the speed of the phone.
  - –There are two kinds of memory (RAM and ROM)
  - –ROM is referred as **storage** = hard disk
  - –RAM is the **memory** where software resides while it is running
- **Camera** – measured in pixel (megapixels)
  - –Good quality photos depends on ISO levels, aperture and speed of autofocus.

Types of Computers

1. Personal computers

- Computers designed for only one person to use at a time

2. Multi-user computers

- Computers designed to serve group of people

Multi-user computers

- Server
- Mainframe
- Supercomputer

Servers

- A **server** is a computer dedicated to providing one or more services to other computers or devices on a network.





Mainframe

- **Mainframe** - computers designed to handle thousands of users



Supercomputer

- **Supercomputer** - are large, fast computers assigned to concentrate on specific scientific tasks.



Computer System



Computer System

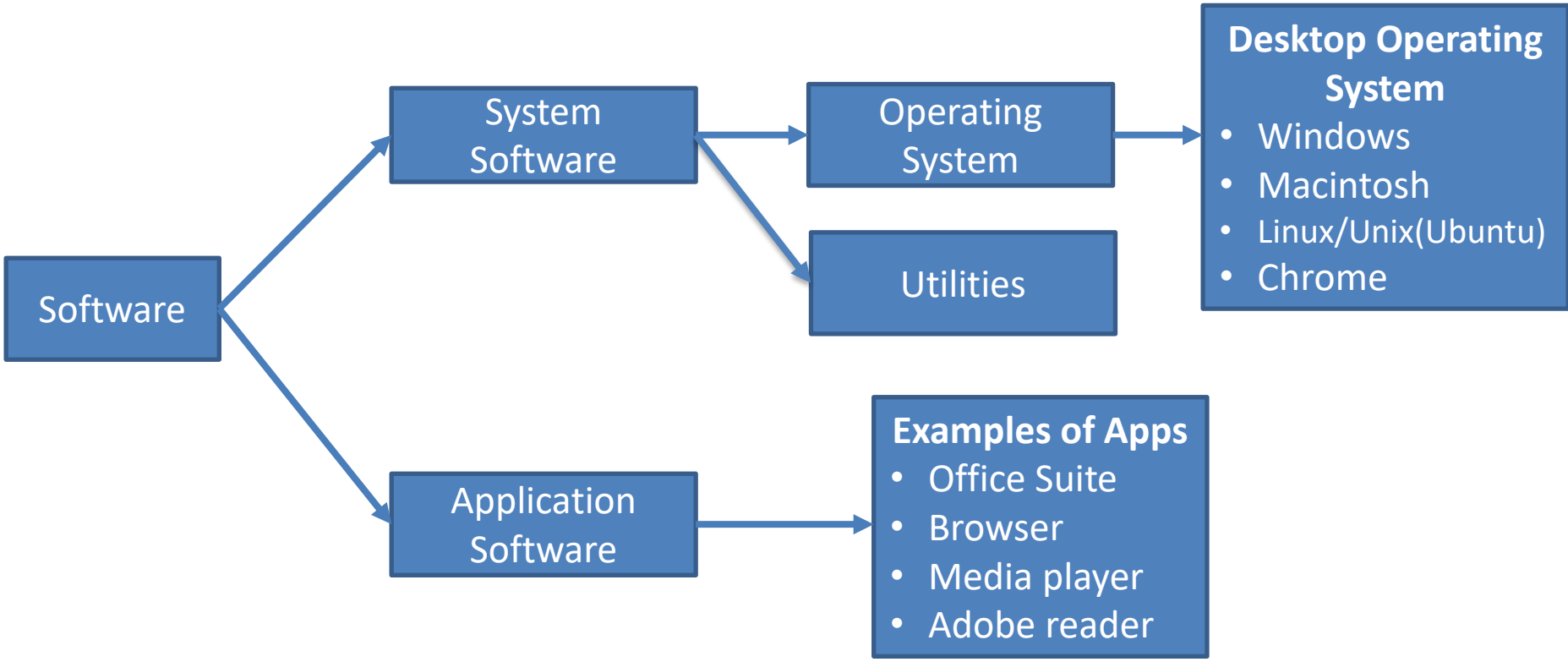


Components of a computer

- **Software** is the set of instructions that tells the computer what to do.

- **Hardware** is the physical component of the computer.

Types of software



Linux OS

- Unix based OS
- **Open source software –** its code is provided for use, modification and redistribution.
- Linux is available in variety of forms known as **distributions.**
- Popular distributions are **Ubuntu, Red Hat, Debian, FreeBSD, Fedora, Android, Chrome OS.**

Role of the OS

i. starting and shutting down a computer or mobile device,
ii. providing a user interface,
iii. managing programs,
iv. managing memory and storage,
v. monitoring performance,
vi. providing file management and other device or media-related tasks

Utilities

- Utilities are applications that perform special functions.
  - Backup
  - Troubleshooting
  - Antivirus
  - Data recovery
  - File compression
  - Uninstall
  - Disk defragmentation

Software Availability

- Retail software
- Custom software
- Web app
- Mobile app
- Open source software
- Shareware
- Freeware

Disadvantages of using computers/Technology

1. Health Risks
2. Cybercrime
3. Public Safety
4. Impact on Labor Force
5. Impact on Environment